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LISF (LIS, LDT, LVT) COMPILATION AND STATUS SUMMARY FOR NOAH-MP MOROCCO STUDY

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Project Directory: /home/mohammad.elaabaribao/lustre/empowermed-ahl6xm8o7mg/users/
mohammad.elaabaribao/NoahMP_Morocco

Summary of Accomplishments:

All three Land Information System Framework (LISF) executables have been successfully configured, patched, compiled, and verified. They are ready to be used for the simulation and comparison steps.

1. COMPILER ENVIRONMENT & MODULES

To compile or run LIS, LDT, or LVT, the following environment modules must be loaded:

```
module purge
module load foss/2024a \
    netCDF-Fortran/4.6.1-gompi-2024a \
    netCDF/4.9.2-gompi-2024a \
    ESMF/8.7.0-foss-2024a \
    JasPer/4.2.4-GCCcore-13.3.0 \
    ecCodes/2.38.3-gompi-2024a \
    HDF5/1.14.5-gompi-2024a \
    CMake/3.29.3-GCCcore-13.3.0
```

Environment variables used during configuration/make:

```
export ESMFMKFILE=/srv/software/easybuild/software/ESMF/8.7.0-foss-2024a/lib/
esmf.mk
```

2. COMPILED EXECUTABLES & PATHS

* LIS Executable:

Path: /home/mohammad.elaabaribao/lustre/empowermed-ahl6xm8o7mg/users/
mohammad.elaabaribao/NoahMP_Morocco/lisf/lis/LIS

* LDT Executable:

Path: /home/mohammad.elaabaribao/lustre/empowermed-ahl6xm8o7mg/users/
mohammad.elaabaribao/NoahMP_Morocco/lisf/ldt/LDT

* LVT Executable:

Path: /home/mohammad.elaabaribao/lustre/empowermed-ahl6xm8o7mg/users/
mohammad.elaabaribao/NoahMP_Morocco/lisf/lvt/LVT

3. CONFIGURATION DETAILS (Generated Settings Files)

The following build configuration files were generated and modified to link EasyBuild libraries:

- * LIS: `lisf/lis/make/configure.lis`
 - Optimization Level: 2 (O2)
 - Endianness: Big Endian
 - Features: ESMF, ecCodes, NetCDF4, HDF5, GRIBAPI
 - Custom linking: netCDF-Fortran and compiler runtime libraries are manually appended to FFLAGS, FFLAGS77, and LDFLAGS.
- * LDT: `lisf/ldt/make/configure.ldt`
 - Optimization Level: 2 (O2)
 - Endianness: Big Endian
 - Features: ESMF, ecCodes, NetCDF4, HDF5
 - Custom linking: Manually linked netCDF-Fortran.
- * LVT: `lisf/lvt/make/configure.lvt`
 - Optimization Level: 2 (O2)
 - Endianness: Big Endian
 - Features: ESMF, ecCodes, NetCDF4, HDF5
 - Custom linking: Manually linked netCDF-Fortran.

4. SOURCE CODE BUG PATCHES (To resolve compile errors with GCC 13/14)

To resolve compilation conflicts between legacy LISF code and modern GCC compilers, two code changes were implemented:

1. LIS Intrinsics conflict:
 - File: `lisf/lis/core/LIS_misc.h`
 - Change: Commented out custom macro redefinitions of standard Fortran intrinsics (`get_command_argument` and `command_argument_count`) to let the compiler use built-in GNU Fortran intrinsics.
2. LDT System intrinsic conflict:
 - File: `lisf/ldt/WSF_OPL/LDT_wsf_oplMod.F90` (around line 555)
 - Change: Commented out the `'external :: system'` statement. This tells the compiler that `'system'` is an intrinsic subprogram rather than an external user-written routine (resolving unresolved external symbols at link-time).

5. HOW TO RECOMPILE IN THE FUTURE (If you clean or update code)

Always load the modules first (see Section 1). Then:

- * To Recompile LIS:
`cd lisf/lis/make`

```
make -f Makefile -j 4
```

```
* To Recompile LDT:  
cd lisf/ldt/make  
make -f Makefile -j 4
```

```
* To Recompile LVT:  
cd lisf/lvt/make  
make -f Makefile -j 4
```

6. NEXT STEPS (Ready for tomorrow!)

Now that the tools are compiled, we are ready to proceed to:

1. Spatial Domain Definition:

- Bounding box defined for the Allal El Fassi Basin:
 - * South: 33.00 N
 - * North: 34.50 N
 - * West: -5.50 W (or 354.50 E)
 - * East: -3.50 W (or 356.50 E)
 - * Resolution: 0.01 degrees (approximately 1 km grid spacing)

2. Parameter Generation (LDT):

- Configure 'ldt.config' with this spatial grid and point to the landcover, soil texture, elevation, and greenness datasets.
- Run LDT (./LDT) to produce the static 'lis_input.d01.nc' file.

3. Forcing and Noah-MP Simulation (LIS):

- Set up meteorological forcing (e.g., GPM/ERA5/MERRA2/NLDAS).
- Configure 'lis.config' for the open-loop simulation.
- Run LIS (./LIS) to generate Noah-MP outputs.

4. Validation and Comparison (LVT):

- Setup LVT configuration ('lvt.config') to read Noah-MP output and compare against satellite observations (soil moisture, LAI, ET).
- Run LVT (./LVT) to compute validation metrics.

7. STEPS FOR 3-MONTH SIMULATION (WITH AND WITHOUT ASSIMILATION)

To run your first 3-month test simulation (e.g., June 1, 2020 to August 31, 2020), follow these steps:

STEP A: Download Meteorological Forcing & Satellite Observations

1. Select a 3-month window: e.g., 2020-06-01 to 2020-08-31.
2. Download/Prepare Met Forcing: MERRA-2, ERA5, or GPM IMERG for the region.
3. Download/Prepare Soil Moisture observations (e.g., SMAP L3 passive or ESA-CCI) for the same period.

STEP B: Generate Model Parameters via LDT

1. Edit 'ldt.config' and specify:
 - The Allal El Fassi Basin grid parameters (0.01 deg resolution).
 - Paths to your static data directories (soils, topography, land cover).
2. Run LDT:
./LDT ldt.config
3. This creates 'lis_input.d01.nc' containing the parameters on your exact 1 km grid.

STEP C: Configure and Run LIS - Open-Loop (OPL) Simulation

1. Create a config file 'lis.config.opl'.
2. Set running mode:
LIS running mode: "retrospective"
Number of data assimilation sources: 0
Data assimilation sources: none
3. Set simulation dates:
Starting year: 2020
Starting month: 06
Starting day: 01
Starting hour: 00
Starting minute: 00
Starting second: 00
Ending year: 2020
Ending month: 08
Ending day: 31
Ending hour: 23
Ending minute: 45
Ending second: 00
4. Specify input parameter file:
LIS parameter table filename: ./lis_input.d01.nc
5. Set output directory:
LIS output directory: ./OUTPUT_OPL
6. Run the Open Loop simulation:
mpirun -np 4 ./LIS lis.config.opl

STEP D: Configure and Run LIS - Data Assimilation (DA) Simulation

1. Create a config file 'lis.config.da'.
2. Set running mode to retrospective with EnKF assimilation:
LIS running mode: "retrospective"
Number of data assimilation sources: 1
Data assimilation sources: "SMAP passive SM" (or "ESA-CCI passive SM")
Data assimilation method: "EnKF"
3. Configure ensemble size and perturbations (required for EnKF):
Number of ensembles: 12 # recommended minimum test size
Perturb met forcing: .true.
Perturb state: .true.
Met forcing perturbation config: forcing_attribs.txt
State perturbation config: state_attribs.txt
4. Set output directory:
LIS output directory: ./OUTPUT_DA
5. Run the Data Assimilation simulation:

```
mpirun -np 12 ./LIS lis.config.da
```

STEP E: Evaluate & Compare using LVT

1. Create 'lvt.config' to compare OUTPUT_OPL vs OUTPUT_DA against independent satellite observations:

- Set runmode: "Verification"
- Under LIS forecast directory, point to: `./OUTPUT\_OPL` and `./OUTPUT\_DA`
- Under Observation directory, point to: your independent satellite observations (e.g. soil moisture, LAI, ET)

2. Run LVT:

```
./LVT lvt.config
```

3. Analyze the generated verification metrics (bias, RMSE, correlation) to evaluate the impact of data assimilation.

STEP F: Custom Python Visualizations

1. A dedicated Python script (`scripts/plot\_da\_comparison.py`) allows you to automatically parse NetCDF files generated by LIS.

2. Ensure you have enabled the `LAI` output parameter in `configs/MODEL\_OUTPUT\_LIST.TBL` (set the first column to 1).

3. Run the analysis script to extract multi-month (e.g., JJA 2020) Soil Moisture, Evapotranspiration, and LAI.

```
python3 scripts/plot_da_comparison.py
```

4. Time-series and spatial maps representing the impacts of assimilation (`DA - OPL`) are exported directly to `experiments/plots/`.

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